

Multimodal Glioma Segmentation and Classification and NLP Analysis of Scholarly Articles to investigate how to improve Segmentation.

Management Report & Proposal

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Executive Summary:

Glioma segmentation and classification models with testing accuracy of 75% and 95% respectively were successfully built. These models were incorporated within a user interface to predict glioma types and where the glioma lies. This user interface also provide the capacity to review reports which can be customized according to business needs. Currently existing reports available relates to expert word analysis and keyword frequency.

Introduction:

It is of paramount interest to detect tumors at early stages in order to provide essential treatment and reduce mortality rate. In particular gliomas being among the most lethal types of neoplasms in people warrants active steps to invite screening and checkups regardless if individuals have no to severe symptoms of glioma. This screening processes should be non-invasive in order to promote awareness and minimal to no complications to persons who are being screened. The current procedure to detect and classify glioma's require for experts to analyze MRI images accounting for possibly edema, enhancing tumor and necrotic core after which another step is required to identify glioma type. This multi-step procedure may at times prove to be laborious for neuro-oncologists or experts to isolate glioma based on the criteria mentioned before and also may be subject to bias, human error or variation of results when analyzed by other experts. Coupling these challenges after isolating these gliomas manually experts are faced with verifying whether glioma is high grade or low grade. It is at this point invasive methods are utilized since distinction in glioma types are not readily visible by the naked eye in many cases. It is important to note that MRI images may be available in various formats t1, t1ce, t2 and flair which may not always show the features readily for humans to intuitively assess.

To resolve the issue of minimizing these laborious steps, removal and minimization of invasive processes, removal of human bias and speed to obtain result. It is necessary to implement machine learning principles that will be capable to analyze images autonomously. The creation of a tool that can handle classification and segmentation will shift the focus away from performing analyses after serious symptoms were observed to analyzing brain scans as a regular checkup procedure. This in itself would improve self-awareness among the populace and improve armamentarium to detect neoplasms in early stages and administering relevant treatments.

Machine Learning can be accomplished by use of convolutional neural networks that produce high accuracy predictions with the flexibility to improve models as data grows. Multiple ways can be used to assess reliability of the tool which includes similarity scores, accuracy, precision and specificity and sensitivity.

Methods

The tool was built by creating a user interface that allows users to upload pngs images that would analyze brain slices and produce predictions of the segmented glioma, the type of glioma present and if no glioma(neoplasm exist). To build prediction models glioma brain data from brats was used for training and validation. Images from BRATS2019 dataset were converted to png files that was comprised of axial, sagittal and coronal planes as well as image types of t1, t2, t1ce, flair and segments.

Segmentation model required aligning brain scans with their associated segments and training the model by labeling the glioma using the segmented images provided by experts. The criteria that was used for segmentation included edema, necrotic core, and enhancing tumor. This model was then evaluated by specific metrics namely accuracy, precision, similarity dice score, specificity and sensitivity.

Classification model required separating slices based on glioma type. All images that were not segmented by an expert were classified as no tumor slice and placed in a unique folder called NO_TUMOR. As for the remaining slices if the original image was already labelled as high grade then those images were stored in a folder a HGG otherwise they labelled low grade they would be stored in a folder called LGG. Storing the images in respective folders created the basis to train classification therefore images were trained to predict three outcomes such as non-tumor image, low grade glioma image or high grade glioma image. The model was primarily evaluated by using accuracy, validation loss and validation accuracy.

Natural Language Processing was done on scholarly articles over the last 10 years to validate authenticity of expert choice to label segmented images based on edema, necrotic core and enhancing tumor. The findings indicate that these terms were referenced in multiple articles over the last 10 years. Although the terms were not heavily contextually used the frequency of use indicates some significant degree of importance. In addition the analysis of the documents convey the most contextually used word and phrases were related to blood vessels and glioma stem cells. After reviewing a variety of papers it was observed that glioma stem cells are responsible for angiogenesis of aberrant blood vessel formation. Therefore it is anticipated that including blood vessel in segmentation of images perhaps may improve segmentation modeling for gliomas.

Segmentation and Classification user interface was built as a dashboard allowing users to upload images and generating glioma segmentation and classification. A unique environment was added to the tool that allows doctors to upload patient folders and review manual segmented compared to predicted segment. In addition reports were added as bar charts for NLP analysis. Each tool function were separated as individual tabs.

Note analysis of images incorporated an algorithm that displays the image with the largest predicted segment first. This is an assumption that the largest predicted glioma segment is the highest priority.

Results

Segmentation and Classification models

All images analyzed by the dashboard will convey performance based on these values.

Metric	Classification	Segmentation
Accuracy	99%	97 %
Validation Accuracy	90%	95%
Dice Score	-	54%
Specificity	-	97%
Sensitivity	-	64%
Precision	-	43%

The models were tested on random images that are subjected to noise. Noise in this case is defined as tissues of the head other than the brain for example nose, eyes, teeth, skull etc. The performance was evaluated against 44 images where the performance³ is denoted below.

Metric	Segmentation	Classification
Accuracy	95%	75%

Area of Opportunity:

- 1) Segmentation model dice score needs improvement so that a greater similarity score can be achieved. This may involve reviewing how training images were processed.
- 2) Segmentation and Classification Models are independent of each other and so business logic may be necessary to indicate which model should be invoked first.
- 3) Only 10% of the training data was used to build classification model due to inaccessibility to compute resources
- 4) Investigation to be done as to which image should be displayed first when performing prediction such as Image with largest segment or image with the greatest probability of high grade or low grade glioma or perhaps a combination of rules.
- 5) Tool can be scaled to minimize latency by considering storage and compute. Some optimization process may include analyzing one patient at a time. Storage images and then analyzing based on specific features of a image instead of all of the images etc.

Conclusion

Leveraging a user interface to capitalize on accessing the predictive capabilities of models has demonstrated that glioma neoplasm diagnosis can be achieved without non-invasive procedures. In addition the tool shows versatility in being able to validate expert input against prediction to ensure the model performs adequately as well as the capacity to display other reports that may be relevant for medical practices.

Proposal

Domain Background:

A glioma is the most common type of brain tumor or central nervous system tumor that occurs in both adults and children and originates in the glial cells that support and surround brain cells such as astrocytes, oligodendrocytes and ependymal cells. These tumors are classified by the World Health Organization (WHO) between grades I – IV where grade IV is the most severe case of Glioblastomas. Severity is measured based on malignancy of the tumor which is constituted by high recurrence rates, worst prognosis and fewest therapeutic options. As a result Glioma Grades I and II are referred to as Low Grade Glioma (LGG) which are slow growing and Glioma grades III and IV are regarded as High Grade Glioma's (HGG) which are fast growing and highly invasive.

The early detection and classification of gliomas are of paramount interest in medicine as this would significantly impact the life expectancy of patients as well as define treatment pathways that are specific to the class or grade of the glioma minimizing complications and or promoting remission. It is particularly important to perform diagnosis via noninvasive methods which prevent exposing patients to complications associated with biopsies such as infections and bleeding of the tumor. Non-invasive methods include MRI and CT scans where the MRI scans are generally more adept in providing images that are more conducive in detecting abnormal tissues in the brain.

In order to detect, isolate and classify gliomas in an appropriate time that would aid in diagnosis and ultimately survival rate for patients living with Glioblastoma (GBM) research has shown how machine learning bridges the gap in segmenting and classifying gliomas through abstract phenotypic analyses of gliomas using Convolutional Neural Networks.

In most documented implementations of segmentation and classifications of Gliomas, MRIs are selected as the preferred image source for machine learning using Convolutional neural networks. Convolutional Neural Networks (CNNs) are promising because they are more reliable than traditional priori or theoretical deductions which are dependent on human experts. On the other hand Deep Learning models do not require preselection and possess the ability to learn which features are most relevant for classification and or prediction (Madeleine M. Shaver et al, 2021). CNNs have provided significant strides in detecting and classifying gliomas however it should be noted that this neural network algorithm requires large amounts of annotated medical datasets which is difficult to obtain also these models are highly dependent on initialization. Resolving these issues typically are compensated for by analyzing 2D versions of the dataset rather than 3D versions, transfer learning and feature extractions. In general when comparing CNNs to traditional human / machine learning classification, contemporary deep

learning models such as CNNs are able to learn when new data is introduced through abstraction of feature importance's as opposed to limitations associated with human errors or predefined features based on expert opinions. The accuracy of CNN in classifying various types of Glioma in general exceeds 90%.

To this end mri scans provides the avenue to non-invasively segment tumors and classify them as high grade and or low grade glioma's.

Objective:

The objective of this project is to build a user interface or dashboard that will segment and classify gliomas in real time by uploading mri scans. In addition analysis of the last ten year corpus will be done to validate the usefulness of segmenting images using necrotic core, edema and enhancing tumor as the basis for building a segmentation model.

Incidents:

Only 12% of all sudden or unexpected death occurs due to glioblastoma's (Riezzo, Irene, et al, May 2, 2013) in other instances late detection of glioma's arise based on symptoms that patients experience and diagnosis provided when survival rate is low. Therefore the detection of these tumors in early stages significantly enhances and or promotes the survival rate of patients.

Data Description:

Structured Data: All images used from training were obtained from kaggle at <https://www.kaggle.com/aryashah2k/brain-tumor-segmentation-brats-2019>. These images are stored as nii files and are separated as High grade and Low grade gliomas where each image was presented in 4 different formats namely flair, t1, t2, t1ce and their respective segments distinguished by experts as edema, enhancing tumor, necrotic core. It is important to note that the nii files also contains three planes for scans such as sagittal, coronal and axial (Each nii file represents 155 slices of the brain in each plane end to end). The total size of the data is 25.74 GB and was compiled over numerous years by various medical institutions.

Unstructured Data - Scholarly articles from Google Scholar, Crossref and Scihub would be analyzed to validate the significance of experts segmenting images using edema, enhancing tumor and necrotic core which would be ultimately used to build segmentation models. In addition statistical analyses will be done to identify if there are any other key features that could be labeled and or included to improve segments provided by experts. Data would be obtained by using Pypaperbot. The total amount of papers obtained from Pypaperbot webscraping is 92 which constitutes 187 Mb.

Data Science Objectives:

The method to gain insights on structured data (nii files of mri scans) is to extract the images as png in order to save as much information about the images and prevent loss. The images will be

resized and saved based on the plane it is associated with (Sagittal, Axial, Coronal). After extraction of the images they will be analysed in terms of valid images taking into account pixel count, sharpness, and blank images and then subsequently each slice will be separated as LGG, HGG or No tumor.

Structured Data (MRI Scans)

- Extraction of images from nii files as PNG to prevent loss
- Resize images appropriately for Segmentation and classification
- Normalize images to minimize differences in image type
- Align segments with respective images appropriately

Unstructured Data (Scholarly Articles)

- Isolate significant and contiguous phrases
- Find most frequent words
- Removal of noise from data extraction

Key Deliverables:

Two models will be built using convolutional neural networks. One of which will be a unet model for segmentation and the other will be a classification model.

The segmentation model will be able to highlight regions of the image that is a glioma and the classification model will be able to distinguish between images as High grade glioma, Low grade glioma and non-tumor.

The final product will be a dashboard that allows users to upload images and or mri scans as png which identifies the segment on the image that is a glioma as well as classify the scan as a high grade, low grade glioma or no tumor. These segmentations and classifications will be analysed as 2d images and also this tool will allow the user to download the classification result as a csv file.

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